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Advanced optimization of drone frame design through the application of generative design techniques and 3D printing technology

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Abstract This study explores the integration of generative design and additive manufacturing (AM) to optimize the structural design of unmanned aerial vehicle (UAV) frames. Carbon fiber-reinforced nylon (PA6-CF) was selected as the primary material, utilizing fused deposition modeling (FDM) for prototyping. The research began with the configuration of UAV frame designs, followed by the application of generative design methodologies to generate over 138 design solutions. These were narrowed down to 35 feasible designs based on mechanical performance criteria, with SS-O11 and NO SS-O15 emerging as the top candidates. The selected designs were subjected to finite element analysis (FEA) for static and dynamic simulations. SS-O11 demonstrated minimal displacement (0.12 mm) under a 1400-gram load, a von Mises stress of 3.28 MPa, and a significant weight reduction of 66.28 % from its initial design. Impact tests showed SS-O11 reaching a peak stress of 152.92 MPa, confirming its structural resilience. The study concludes that integrating generative design with AM effectively produces lightweight, strong UAV frames, offering significant contributions to UAV structure optimization.

1. Introduction

Unmanned aerial vehicles (UAV), commonly referred to as drones, are aircraft operated autonomously or remotely. Without the need for onboard pilots, UAV can carry out missions with high efficiency, making them valuable across various sectors [1] such as defense [2], agriculture [3], healthcare [4], logistics and passenger transport [5].

Technological advancements and diverse UAV applications have significantly driven the growth of the global drone market. The Drone Market Report 2021 by Drone Industry Insights predicts an annual growth rate of 9.4 % until 2026 [6], while Levitate Capital projects the drone market to reach USD 90 billion by 2030 [7]. In Indonesia, the drone market potential is projected to reach USD 93 million or IDR 1.47 trillion by 2028 [8].

The aerospace industry continues to develop superior materials and structural designs, aiming to balance lightness and strength [9]. Composite materials have become the primary choice over heavier metal alloys in aircraft structures [10]. Optimization methods such as topology optimization and generative design have become research focal points in achieving optimal structural designs.

Generative design offers the potential to create lightweight and strong structures by considering various factors. Design optimization can be based on load distribution and stress, and the results can be directly processed for fabrication without the need for iteration.

Generative design has been explored in the production of components such as UAV landing gear [11], electrical generator structures [12], optimization of bionic prosthesis finger joints [13], and robotic exoskeleton thumb joints [14].

Additive manufacturing (AM), or 3D printing, has become a major trend in UAV structure manufacturing due to its ability to create complex [15] and lightweight geometries [16]. UAV structural weight can be significantly reduced using internal lattice structures [17] and numerical optimization tools like generative design. This technology enables weight reduction, shorter production times, and lower production costs [18], aligning with the principles of design for manufacture and assembly (DFMA) [19], and is thus considered an ideal complement to topology optimization and generative design methods.

Previous research has demonstrated advancements in UAV technology and design optimization. However, there is a gap in integrating additive manufacturing technology and generative design methods in UAV structure fabrication. This study aims to bridge this gap by combining these approaches to create more high-performance UAV structures. Given the importance of developing optimal structural designs for UAV, this research is relevant and has the potential to significantly contribute to improving efficiency, performance, and innovation in the UAV industry and related industries in the future.

2. Configurable design

To design a drone frame, the maximum take-off weight and maximum thrust-to-weight ratio must be defined first. Then, the geometric dimensions can be determined through the maximum thrust force of each propeller calculated using Eq. (1).

$$F = \frac{M_{\max} \times P}{n} \tag{1}$$

Where F is the maximum thrust force of the rotor, M_{max} represents the maximum take-off mass, P is the maximum thrust-to-weight ratio, and n is related to load requirements and design motivation.

The relationship between geometry, wheelbase (L), and maximum propeller radius ($rmax$) is shown in Fig. 1, with the mathematical Eq. (2).

$$\sin\theta = \frac{\gamma_{\max}}{L/2} \tag{2}$$

And

$$\theta = \frac{360}{2n}, n = 4,6,8 \tag{3}$$

To create a compact UAV frame, the following requirement must be met

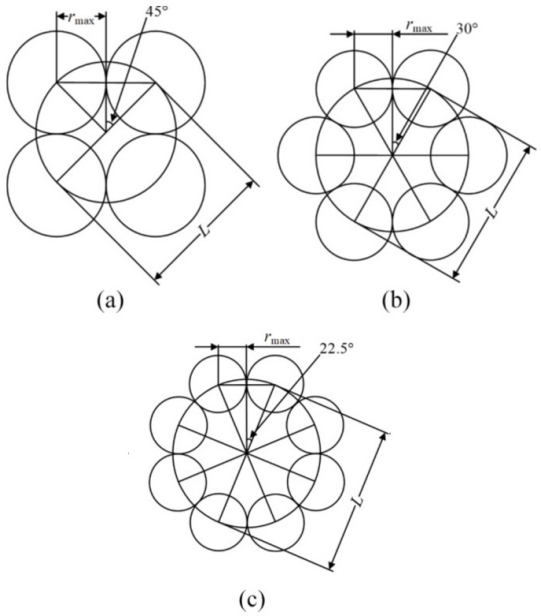


Fig. 1. Types of propellers: (a) quadrotor; (b) hexarotor; (c) eight-rotor [20].

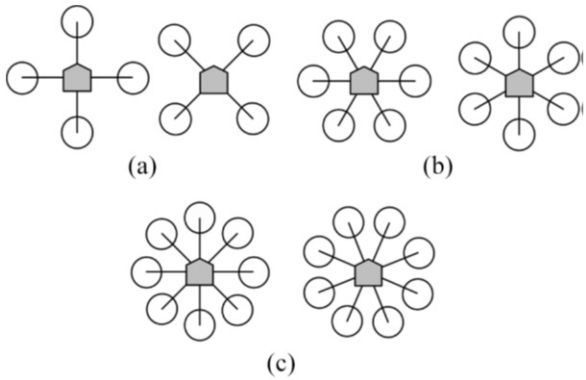


Fig. 2. Layout of rotary-wing UAV: (a) quadrotor layout; (b) hexarotor layout; (c) eight-rotor aircraft layout [20].

$$\gamma_{\max} - \gamma_p > 0.05\gamma_{\max} \tag{4}$$

where r_p is the propeller radius, determined based on the mass and load of the UAV frame. The minimum wheelbase can then be determined using Eq. (2).

Multi-rotor UAV, which are commonly used, are categorized into several variants based on the number of rotors used, such as quadrotor, hexarotor, and octarotor UAV. Additionally, frame layouts are often categorized into two main types: "+" and "X" for multirotor UAV, as shown in Figs. 1 and 2. Therefore, it is important to have a design that can be adapted to specific configurations, considering the frame size parameters that will set initial constraints in the design process.

3. Material and method

3.1 Material

Table 1. Mechanical properties data of carbon fiber nylon provided by the supplier.

Name	Orientation		
	X	Y	Z
Strength (MPa)	7453	7453	4354
Young's modulus (MPa)	105	105	67.7
Elongation at break (%)	3	3	2.5
Bending modulus (MPa)	8339	8339	
Bending strength (MPa)	169	169	
Charpy impact strength (MPa)	13.34	13.34	
Density (g/cm^3)	1.171		

Carbon fiber-reinforced nylon (PolyMideTM PA6-CF) filament was selected as the primary material for the UAV. PA6-CF enables the production of strong and lightweight components, making it ideal for applications requiring high structural strength while maintaining low weight, particularly in UAV components [21]. The use of FDM technology has also proven efficient in terms of cost and time, facilitating prototype development and reducing the cost and production time of UAV components [22, 23].

The carbon fibers are supplied in spools of carbon fiber composite filament. This composite filament is composed of multiple strands of carbon fibers embedded in a nylon 6 matrix. The mechanical properties of the carbon fiber composite in the fiber direction, as reported by the supplier, are shown in Table 1.

3.2 Method

The fused deposition modeling (FDM) method was chosen for producing UAV due to its cost-effectiveness and ease of use for initial prototyping. FDM enables the creation of strong and lightweight structures at a low cost, facilitating rapid prototyping and design adjustments [24, 25]. Several thermoplastic polymers have been successfully applied in FDM, including polyamide (PA) [26], polylactic acid (PLA) [27], and thermoplastic polyurethane (TPU) [28], among others.

The drone frame model was printed using the Bambu Lab X1-carbon 3D printer. The print setting is as shown in Table 2. The preparation process before 3D printing with PA6-CF material involves drying the filament in a filament dryer for 7 hours at 70 °C. During the 3D printing process, the filament remains in the dryer to minimize the risk of hygroscopic absorption.

3.3 Generative design

The initial stage in designing a quadcopter frame begins with gathering supporting data and modelling the quadcopter frame using the "DJI PHANTOM 4 RTK" frame model as a reference. The next step is to establish generative design parameters, including setting exploration design goals, determining geometry, load cases, objective, manufacturing methods, and materials to be used [29].

Table 2. Print settings of Bambu Lab X1-carbon 3D printing.

Nozzle temperature (°C)	280
Bed temperature (°C)	80
Ambient (°C)	40-50
Flow ratio	0.799
Fan	Off
Retraction speed (mm)	2.8
Retraction distance (mm/s)	40
Pressure advanced (s)	0.06

Once all input parameters are available, the design exploration process begins. The software generates alternative design solutions based on the established parameters. During the design selection phase, visualization tools provided by generative design are used to identify the best design based on performance criteria. The final stage is design validation, which involves finite element simulation analysis to minimize errors before the prototype fabrication process. Additionally, the validation results will be printed using three-dimensional (3D) printing technology.

3.4 Generative design scheme

The initial structure parameters and other components are determined by referencing the DJI PHANTOM 4 RTK UAV. This drone is specifically engineered for surveying, mapping, and other tasks demanding precise aerial imaging. With advanced flight control systems, high-resolution cameras, and GPS capabilities, it's tailored for professional use in agriculture, construction, and land surveying.

Components like the flight battery and camera are labeled PH4-5870 and DJI gimbal camera, respectively. Table 3 outlines the UAV key parameters, including rotor count, maximum takeoff mass, thrust-to-weight ratio, and wheelbase. The initial design domain of the UAV frame, depicted in Fig. 3, weighs 420 grams, divided into 396.668 tetahedra elements.

Each propeller of the DJI Phantom 4 RTK generates a vertical upward force of 8 newtons (N) as per the motor specifications. For this study, the force is varied to three different values: 8 N, 10 N, and 15 N. Additionally, a torque of 13 newton millimeters (Nmm) is applied at the motor arm section. A distributed mass of 1.400 kilograms (kg) is used to simulate the thrust, torque, and mass of the motor installed on the unmanned aerial vehicle (UAV). The model is fixed at a single point, as illustrated in Fig. 3.

The exploration of the drone frame design is conducted with two different configurations to obtain optimized results. The first configuration involves optimization without an initial design as the optimization guidance (no starting shape, NO SS). The second configuration involves optimization with a predetermined initial design (starting shape, SS), as seen in Fig. 4.

The drone frame design is explored using two different con-

Table 3. Basic parameters of the UAV.

Name	Detail parameter
Number of rotors (n)	4
Maximum takeoff mass (M_{max})	1.3 kg
Max thrust-to-weight ratio (P)	2.3
Propeller	DJI 9455 S
Motor	DJI 2312 S
Wheelbase	350 mm
Propeller diameter	240 mm
Propeller mass	0.012 kg
Motor mass	0.0059 kg
Electronic equipment mass	0.857 kg
Frame mass	> 250 kg
Battery installation size	150 mm × 113 mm × 70 mm

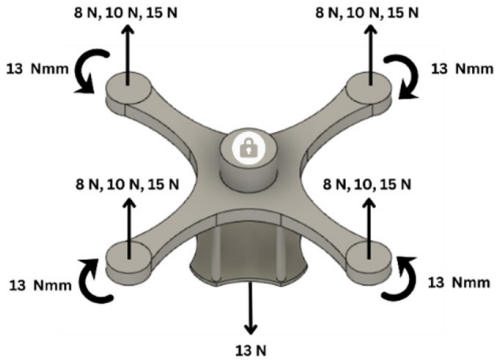
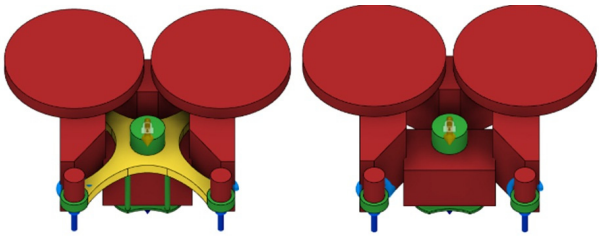


Fig. 3. Initial design of frame drone.



(a) (b)

Fig. 4. Generative design schemes: (a) starting shape (SS); (b) no starting shape (NO SS).

figurations to achieve optimized results. The first configuration involves optimization without a predefined initial design (no starting shape, NO SS). The second configuration incorporates a predetermined initial design (starting shape, SS), as illustrated in Fig. 4.

In generative design, determining the geometry requires identifying areas that must remain intact (preserve regions) and areas that should be avoided (obstacle regions) during the optimization process. Preserve regions are essential parts of the structure that must not be altered, while obstacle regions are areas where material should be avoided. In Fig. 4, red indicates obstacle regions, and green indicates preserve regions.

Table 4. Numbers of solution by material.

Material	Completed	Converged	Failed
CFRN	38	12	10
ABS plastic	33	17	10
Aluminum	18	0	0
Total	89	29	20

Table 5. Numbers of solution by manufacturing method.

Manufacture	Completed	Converged	Failed
Unrestricted	5	5	2
Additive	36	24	10
Casting	48	0	8
Total	89	29	20

This step is crucial as it directly influences the final shape of the simulation results. By defining preserve regions, we ensure that critical components and structural pathways remain undisturbed during optimization. Obstacle regions help avoid interference with other components and adhere to manufacturing constraints. This careful definition is vital for producing a design that is both optimal and functional, and that facilitates manufacturing through 3D printing. Properly defining these regions ensures the resulting design is strong, lightweight, and manufacturable.

Despite achieving promising results from the generative design, further optimization is needed due to issues with material redundancy. Manual refinements using CAD software, such as the sculpt feature, are used to replicate the suggested shapes. Transforming the generative design into a manufacturable product presents challenges, but advancements in 3D printing technology enable the production of complex geometries with high precision. Adjusting dimensions post-generative design ensures optimal structural thickness, enhancing the manufacturability of the UAV frame through 3D printing.

4. Results

4.1 Generative design

The generative design produce over 138 design solutions, divided into three categories: completed, failed and converged. Completed designs have undergone the generative design process successfully but do not meet the performance criteria or have errors in one or more geometries. Conversely, converged designs meet the minimum criteria for mechanical performance. The outcomes of the generative design exploration are summarized in Tables 4 and 5, categorized by materials and manufacturing methods. Additive manufacturing design solutions faced constraints such as manufacturing orientation direction, minimum thickness, and overhang angle. However, unrestricted manufacturing methods allowed for more varied

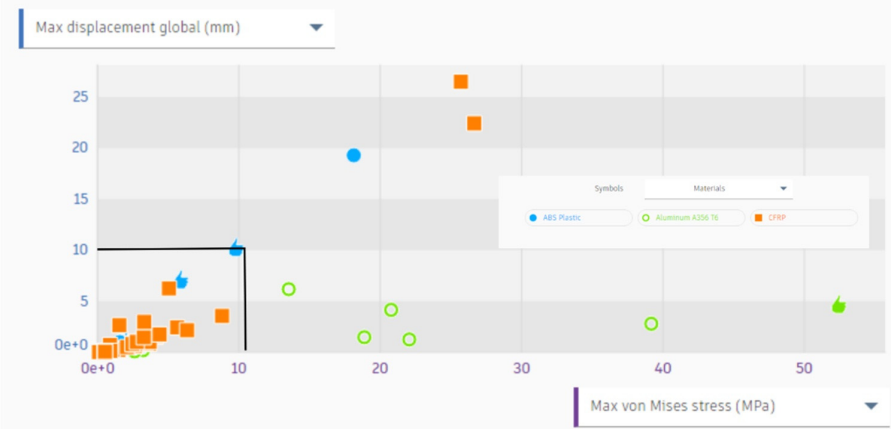


Fig. 5. Scatter plot graph max. displacement & max von Mises stress.

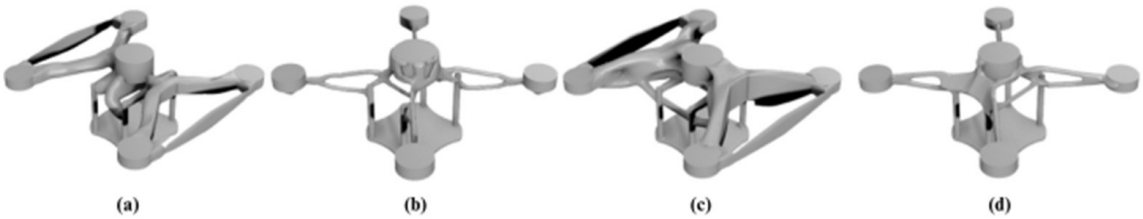


Fig. 6. Generative design schemes: (a) unsymmetrical geometry; (b) poor surface quality; (c) geometry that exceeds the starting shape limit; (d) irregularly formed design geometry.

geometries in the design solutions.

These results were visually represented on a scatter plot graph generated by Autodesk Fusion 360 - generative design (Fig. 5). This graph facilitates users in mapping the mechanical performance of each design within the constraints of mechanical performance criteria, particularly those with converged solution status. The mechanical performance criteria include a maximum displacement of 1 mm and a maximum von Mises stress under 4 MPa. Designs meeting these criteria are deemed successful outcomes, enabling users to identify and select optimal design solutions based on mechanical performance.

4.2 Design selection

Within the scope of the research, there are 118 designs that have reached converged and completed status. However, it is important to observe and consider the visual aesthetics of each design. Generative design also provides recommendation values based on the mechanical performance of each design. The recommendation value feature aims to assist designers in making decisions on the final design to proceed to the next stage. It is hoped that the analysis process can select designs that have feasibility in terms of mechanical performance factors and visual aesthetics.

Consideration factors used include design neatness and feasibility for the manufacturing process. Fig. 6 shows visually identifiable defect design criteria based on (a) unsymmetrical geometry, (b) poor surface quality, (c) geometry that exceeds the starting shape limit, (d) irregularly formed design geometry.

Formed due to adjustments to the limitations of manufacturing methods used as parameters in generative design. When the main parameters of mechanical criteria are consistently maintained in each outcome, sometimes the software fails to produce geometric details in the design. After careful observation, several outcomes show a better level of feasibility in terms of visual aesthetics or a design with feasible criteria that can be identified based on good surface quality, irregular but symmetrical design geometry, and geometry that does not exceed the limit of the initial shape [29].

Data on the mechanical performance of best designs and the recommendation values of each category can be found in Tables 6 and 7. Based on the recommendation value, six designs (Fig. 7) with the highest recommendation values from each category were chosen from a total of 35 designs. Furthermore, design validation through finite element analysis (FEA) simulation in static stress study with input parameters is the same as the exploration scheme of generative design in Fig. 3.

5. Discussion

Fig. 8 illustrates the von Mises stress distribution across various load scenarios for six different materials. Generally, von Mises stress increases with load complexity or intensity in all models. Model NO SS-O1 exhibits the highest stress increase across all scenarios, indicating the lowest resistance to

Table 6. Best design with starting shape.

Name	Load cases (N)	Rec (%)	Material	Manufacturing method	Mass (kg)	Max von Mises stress (MPa)	Min factor of safety	Max displacement global (mm)	First mode frequency (Hz)
SS-O11	8	91.61	CFRP	Unrestricted	0.2096	1.0681	46.8115	0.2468	40.4675
SS-O1	15	91.45	ABS Plastic	Unrestricted	0.1967	2.0635	9.6925	0.9724	31.1264
SS-O7	10	71.53	ABS Plastic	Additive	0.1918	1.6207	12.3402	0.7061	28.9254

Table 7. Best design without starting shape.

Name	Load cases (N)	Rec (%)	Material	Manufacturing method	Mass (kg)	Max von Mises stress (MPa)	Min factor of safety	Max displacement global (mm)	First mode frequency (Hz)
NO SS-O1	10	91.74	ABS Plastic	Unrestricted	0.2018	0.3348	59.7325	0.0783	21.7727
NO SS-O11	10	89.32	CFRP	Unrestricted	0.2129	3.3058	15.1248	2.9569	9.4964
NO SS-O15	10	71.74	CFRP	Additive	0.2036	0.5277	94.7452	0.0429	25.3308

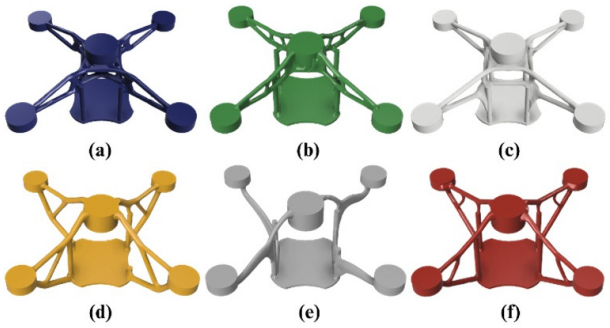


Fig. 7. Best design: (a) SS-O1; (b) SS-O7; (c) NO SS-O11; (d) NO SS-O1; (e) NO SS-O11; (f) NO SS-O15.

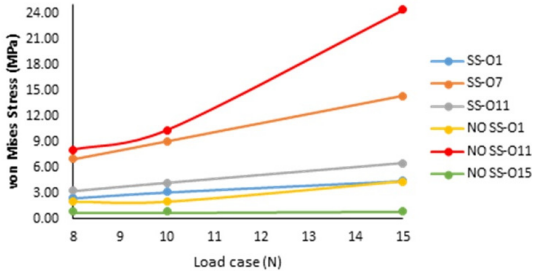


Fig. 8. Comparison of stress in static simulation.

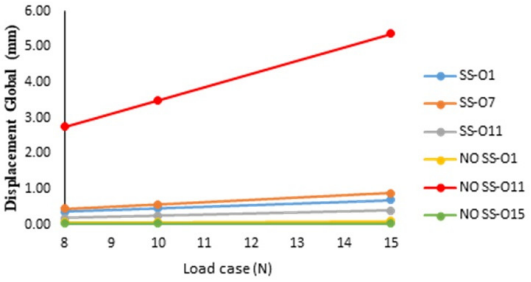


Fig. 9. Comparison of displacement in static simulation.

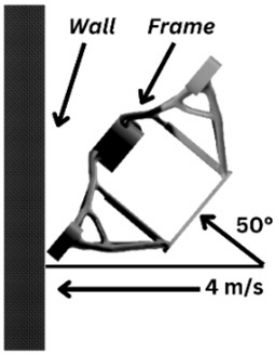


Fig. 10. Load case impact test.

increased load. Model SS-O7 also shows a significant stress increase. Conversely, materials SS-O1, SS-O11, NO SS-O11, and NO SS-O15 display lower and more stable stress increases. Fig. 9 depicts global displacement in structural analysis under varying loading conditions. Models SS-01, SS-07, SS-011, and SS-015 show minimal global displacement increase with added load. However, model NO-SS-011 demonstrates a significant rise in global displacement with increased load, indicating a greater load distribution in this model compared to others. Based on these observations, designs SS-011 and NO-SS-15 were selected as the best results from the static simulation, representing two exploration schemes: starting shape and no starting shape.

The dynamic simulation involved impact tests to evaluate the performance and durability of the drone structure during crashes. The reference model used is the DJI Phantom 4 RTK with a maximum impact speed of 4 m/s. During the impact test simulation, both drones were subjected to a collision at 4 m/s with an angle of 50 degrees (Fig. 10). The impact and falling surfaces were idealized as hard surfaces (stone) to simulate the worst-case scenario. Explicit dynamic analysis was performed with a time step of 0.05 s.

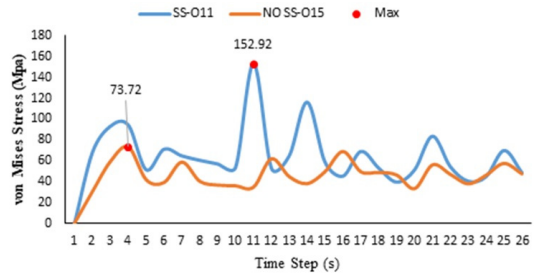


Fig. 11. Comparison of stress in impact test simulation.

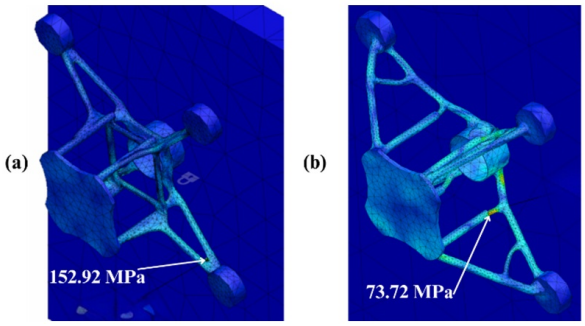


Fig. 12. Impact test: (a) SS-O11; (b) NO SS-O15.

The impact test results on the SS-O11 frame show an increase in von Mises stress over time after the impact occurs. The stress peaks around time step 10 with a value of 152.92 MPa. Similarly, the NO SS-O15 frame exhibits a similar pattern, with von Mises stress peaking around time step 3 with a value of 73.72 MPa. Fig. 11 illustrates the stress fluctuations, which appear on the graph as peaks and valleys. This may be due to the natural frequency resonance of the frame following the impact, uneven stress distribution due to the 50-degree impact angle, and stress exceeding the yield strength during the simulation. The amplitude reduction in stress reflects the damped vibration energy, which then increases again as the impact energy continues to affect the frame [30].

In the simulation (Fig. 12), the smallest values for yield strength and tensile strength were used 50 MPa and 67.7 MPa, respectively due to software limitations, as referenced in Table 1. However, testing by Ref. [31] shows that carbon fiber-reinforced nylon has a yield strength of 161 MPa and a tensile strength of 182.59 MPa. Both static and dynamic simulations suggest the designs are safe, as stress levels remain below the tensile strength limit. Experimental testing is necessary to confirm whether the designs remain safe and do not undergo plastic deformation under crash conditions. The SS-011 design was selected for experimental testing. Fig. 13 details the test specimen and setup for load testing. After fabrication, the UAV is clamped onto a table, and a gradually increasing mass ranging from 200 grams to 1400 grams is added to a suspended bag attached to the clamp to achieve the appropriate displacement. Vertical displacement of the UAV is measured using a displacement gauge. Due to dimensional limitations of the Bambulab X1 carbon 3D printer, the design size was reduced

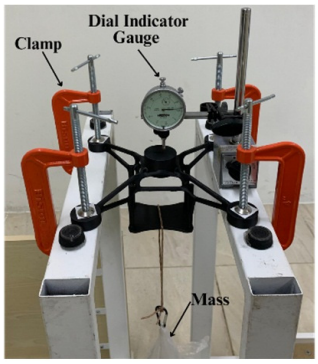


Fig. 13. Experimental testing.



Fig. 14. 3D printing of SS-O11.

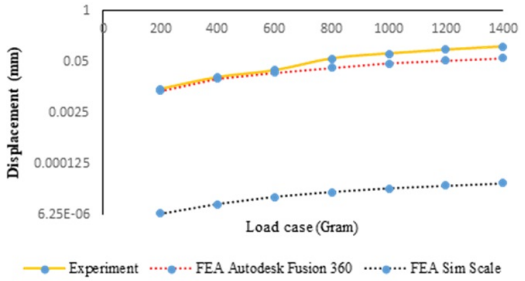


Fig. 15. Comparison of vertical displacement between experimental results and FEA results.

by 10 %, resulting in a initial weight of 344 grams. After printing, the SS-011 design weighed 116 grams, representing a mass reduction of 66.28 % from the initial shape's weight of 344 grams (Fig. 14).

Experimental testing revealed a significant difference between the experimental results and the two FEA (finite element analysis) methods used. The experimental results were higher compared to the FEA, particularly under a static load of 1400 grams, which resulted in a vertical displacement of 0.12 mm (Fig. 15). Nonetheless, this displacement is very low, indicating that the 3D printing result using FDM is quite good.

6. Conclusion

This study highlights the effectiveness of integrating generative design and additive manufacturing to create optimal UAV structures. The generative design method has proven effective in producing lightweight and optimally stiff drone frames. This process yielded 138 initial designs, narrowed down to 118

within research limits, and further refined to 35 feasible designs, with SS-O11 and NO SS-O15 emerging as the top choices. SS-O11, constructed from carbon fiber reinforced polymer (CFRP), exhibits minimal displacement of 0.12 mm under a 1400-gram load and weighs 116 grams, representing a 66.28 % reduction from its initial 344 grams, while meeting stiffness requirements. In static simulations, SS-O11 achieves a factor of safety of 15, with a global displacement of 0.18 mm and von Mises stress of 3.28 MPa. Similarly, NO SS-O15 demonstrates robust mechanical performance with a factor of safety of 15, von Mises stress of 0.83 MPa, and displacement of 0.03 mm. Impact tests reveal SS-O11 reaching a peak stress of 152.92 MPa and global displacement of 163.71 mm, while NO SS-O15 reaches 73.72 MPa stress and 163.64 mm displacement, highlighting their resilience in real-world scenarios.

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References

[1] M. Ayamga, S. Akaba and A. A. Nyaaba, Multifaceted applicability of drones: a review, *Technol Forecast Soc Change*, 167 (2021) 120677.

[2] P. Mahadevan, The military utility of drones, *CSS Analyses in Security Policy*, 78 (2010).

[3] T. Xie, J. Lie, C. Yang, Z. Jiang, Y. Chen, L. Guo and J. Zhang, Crop height estimation based on UAV images: methods, errors, and strategies, *Computer and Electronics Agriculture*, 185 (2021) 106155.

[4] E. Ackerman and E. Strickland, Medical delivery drones take flight in east africa, *IEEE Spectrum*, 55 (1) (2018) 34-35.

[5] R. Kellermann, T. Biehle and L. Fischer, Drones for parcel and passenger transportation: a literature review, *Transportation Research Interdisciplinary Perspectives*, 4 (2020) 100088.

[6] Drone Industry Insight, *Droneii, Press Release - Global Drone Market Growing Rapidly to US$ 41 Billion by 2026 Despite Pandemic, Drone Market Report*, Drone Industry Insight, Available: https:// droneii.com/latest-press-release, Accessed on Jan. 24, 2024.

[7] Levitate Capital, *The Future of the Drone Economy, Dec. 2020*, Levitate Capital, Available: https://levitatecap.com/white-paper/, Accessed on Jan. 24, 2024.

[8] T. K. Yunianto, Potensi pasar drone ri diproyeksikan capai us$ 93 juta, *Marketeers*, Available: https://www.marketeers.com/potensi-pasar-drone-ri-diproyeksi kan-capai-us-93-juta/, Accessed on Apr. 26, 2024.

[9] R. N. Yancey, Challenges, opportunities, and perspectives on lightweight composite structures: aerospace versus automotive, *Lightweight Composite Structures in Transport: Design, Manufacturing, Analysis and Performance*, Woodhead Publishing (2016) 35-52.

[10] P. D. Mangalgiri, Composite materials for aerospace applications, *Bulletin of Materials Science*, 22 (3) (1999) 657-664.

[11] A. Gupta, V. Soni, D. Shah and A. Lakdawala, Generative design of main landing gear for a remote-controlled aircraft, *Material Today Proceedings* (2023).

[12] D. Gonzalez-Delgado, P. Jaen-Sola and E. Oterkus, Design and optimization of multi-MW offshore direct-drive wind turbine electrical generator structures using generative design techniques, *Ocean Engineering*, 280 (2023) 114417.

[13] A. Triono, M. Darsin, A. Fathurrahman, S. Mulyadi and N. Ilminnafik, Optimization of the structure of bionic finger segment prosthesis using generative design, *F1000Research*, 11 (2022) 613.

[14] S. Chandika, *Optimasi Desain Tangan Robot Exoskeleton Ruas Jari Jempol Dengan Penerapan Metode Generative Design*, Fak. Teknik UNEJ (2023) Available: https://repository.unej.ac.id/handle/123456789/173/browse?type=author&value=CHANDIKA%2C+Shendy, Accessed on Jan. 23, 2024.

[15] G. D. Goh, S. Agarwala, G. L. Goh, V. Dikshit, S. L. Sing and W. Y. Yeong, Additive manufacturing in unmanned aerial vehicles (UAV): challenges and potential, *Aerospace Science Technology*, 63 (2017) 140-151.

[16] C. Ferro, R. Grassi, C. Seclì and P. Maggiore, Additive manufacturing offers new opportunities in UAV research, *Procedia CIRP*, 41 (2016) 1004-1010, doi: 10.1016/J.PROCIR.2015.12.104.

[17] T. Han, D. Qi, J. Ma and C. Sun, Generative design and mechanical properties of the lattice structures for tensile and compressive loading conditions fabricated by selective laser melting, *Mechanics of Materials*, 188 (2024) 104840.

[18] M. K. Thompson et al., Design for additive manufacturing: trends, opportunities, considerations, and constraints, *CIRP Annals*, 65 (2) (2016) 737-760.

[19] G. Boothroyd, Design for manufacture and assembly: the boothroyd-dewhurst experience, *Design for X*, Springer (1996) 19-40.

[20] H. Guo, M. Li, P. Sun, C. Zhao, W. Zuo and X. Li, Lightweight and maintainable rotary-wing UAV frame from configurable design to detailed design, *Advances in Mechanical Engineering*, 13 (7) (2021).

[21] A. M. Almushaikeh et al., Manufacturing of carbon fiber reinforced thermoplastics and its recovery of carbon fiber: a review, *Polymer Testing*, 122 (2023) 108029.

[22] S. D. Shelare, K. R. Aglawe and P. B. Khope, Computer aided modeling and finite element analysis of 3-D printed drone, *Material Today Proceedings*, 47 (2021) 3375-3379, doi: 10.1016/J.MATPR.2021.07.162.

[23] F. Ning, W. Cong, J. Qiu, J. Wei and S. Wang, Additive manufacturing of carbon fiber reinforced thermoplastic composites using fused deposition modeling, *Composites Part B: Engineering*, 80 (2015) 369-378.

[24] S. M. Zaharia, L. S. Pascariu, L. A. Chicos, G. R. Buican, M. A. Pop, C. Lancia and V. M. Stamate, Material extrusion addi-

tive manufacturing of the composite UAV used for search-and-rescue missions, *Drones,* 7 (10) (2023) 602.
[25] D. Kumar, *Simplifying UAV Design Through Composites and Additive Manufacturing*, VECROS, Available: https://www.linkedin.com/pulse/simplifying-uav-design-through-composites-additive-manufacturing/, Accessed on Jan. 20, 2024.
[26] S. Arthanari et al., Laser surface polishing of 3D printed polylactic acid (PLA) with different levels of absorption, *J. Manufacturing Process*, 98 (2023) 265-276.
[27] S. Liu, H. Wu, B. Chao, W. Fu, K. Deng and Y. Li, Study on FDM preparation and properties of RGO/Ni/PLA/TPU composite materials, *Journal of Alloys and Compounds*, 968 (2023) 172119.
[28] M. Qian, X. Xu, Z. Qin and S. Yan, Silicon carbide whiskers enhance mechanical and anti-wear properties of PA6 towards potential applications in aerospace and automobile fields, *Composite Part B: Engineering*, 175 (2019) 107096.
[29] F. Buonamici, M. Carfagni, R. Furferi, Y. Volpe and L. Governi, Generative design: An explorative study, *Computer-Aided Design & Applications*, 18 (1) (2020) 144-155, doi: 10.14733/CADAPS.2021.144-155.
[30] T. Al-Hababi, M. Cao, B. Saleh, N. F. Alkayem and H. Xu, A critical review of nonlinear damping identification in structural dynamics: methods, applications, and challenges, *Sensors (Basel)*, 20 (24) (2020) 1-41.
[31] M. Mohammadizadeh and I. Fidan, Tensile performance of 3D-printed continuous fiber-reinforced nylon composites, *Journal of Manufacturing and Materials Processing 2021*, 5 (3) (2020) 68.



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